Dynamical Oceanography

Problem Set 3

Problem 1 is due before spring break. Either give it to me in class or email it to me. The due date for the other problems will be announced later.

1. For a two-layer wind-driven quasi-geostrophic model, obtain an expression for the minimum strength of the wind that leads to closed contours of $\overline{q} = \beta y + F\overline{\psi}$. In particular, for the solution illustrated in Fig. 14.15 of AOFD, at what values of A do closed contours first form?

   Show that the streamfunction and potential vorticity are continuous across the boundary between the closed region (the 'pool') and the shadow zone.

2. *(a)* Consider an ocean with a flat surface, forced only by surface fluxes. Suppose that the net fluxes at the surface fall from about $+100\,\mathrm{W\,m^{-2}}$ at the equator to $-100\,\mathrm{W\,m^{-2}}$ at the pole, and that associated heating $\dot{Q}$ varies in vertical with an e-folding scale of 10 metres with a volume integral that is zero. Is such a forcing sufficient to account for the observed dissipation of kinetic energy in the ocean, which is about $1\,\mathrm{kW\,km^{-3}}$?

   (b) Now let a wind blow over the ocean, providing an average stress of $0.05\,\mathrm{N\,m^{-2}}$. Estimate the kinetic energy dissipation that this will lead to.

   (c) Estimate the internal rate of heating (in degrees per day or similar) associated with a kinetic energy dissipation rate of $1\,\mathrm{kW\,km^{-3}}$?

3. Consider a model of sideways convection in which the boundary condition at the top is the relaxation, or Haney, condition $b_z = A(b^*(y) - b)$ where A (a constant) and $b^*(y)$ are given, b is the buoyancy, proportional to the temperature, the other boundaries are insulating, and the flow is statistically steady.

   (a) Is it still the case that on average the fluid is heated (i.e., there is a heat flux into the fluid through the upper surface) where it is already warm? If so, how may this be reconciled with the intuition that if the ocean surface is anomalously warm it will cool by way of a heat flux from the ocean to the atmosphere?

   (b) Show that if $A < 0$ a statistically steady state cannot be reached.

   (c) Suppose that b^* varies monotonically with latitude. Show that if κ is non-zero, the average surface temperature gradient will be less than that of b^*.